How Time Flies: Emotional and Sonic Change in Music, 1990–2022

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***Abstract*—How has the produced music changed over the last three decades? We analyze the emotional and sonic change in produced music from 1990 to 2022 using a dataset from Spotify which includes 489,761 songs, their lyrics and sonic features. A Spark pipeline and BERT lyrics-emotion classification, K-Means valence and energy era discovery, and Mann–Kendall trend testing and BIC/Pettitt change-point detection have been made. Over 33 years, we find that from 1990 to 2022, valence (22%), acousticness (29%), and duration has decreased, while energy (18%), tempo (+4.9 BPM), and loudness (+3.7 dB) have increased. The lyrics contain more words about loneliness. The dominant turning point is around 2000, which is consistent across all methods.**

***Index Terms*—music information retrieval, emotion classification, BERT, Apache Spark, trend analysis, change-point detection**

I. Introduction

We humans evolve with time and what it brings. Music has been a way for humans to express themselves. This study aims to look at the change of sonic features and lyric emotions in the music spanned over the years. We aim to use a BERT emotion model on the lyrics of 489,761 songs from 1990 to 2022, which extracts the probability of the lyrics' dominant emotion out of love, joy, anger, sadness and fear. We also analyze the audio features of the songs, including valence, energy, loudness, acousticness, tempo and duration. We then analyze the trends of these features over time and detect change points in the data. We also use K-Means clustering to discover eras in the music based on valence and energy.

A. Problem Statement

We ask whether the emotional and sonic character of popular music changed measurably between 1990 and 2022:

- **RQ1.** Do lyric emotions show significant patterns over time?
- **RQ2.** Do audio features (valence, energy, loudness, tempo, . . .) show significant trends?
- **RQ3.** Does valence and energy clustering reveal discrete eras?

B. Contributions

- **Mood and lyric emotion at scale.** Sound (ten audio features) and words (BERT lyric emotion) measured across 489,761 songs over three decades.
- **Validated with statistics.** Mann–Kendall, Sen's slope, Benjamini–Hochberg FDR, change-point detection, and Pettitt's test implemented from scratch and cross-checked against SciPy, statsmodels, and ruptures.
- **Comparing results for yearly differences.** Break years from all analyses synthesized into one timeline, converging on a dominant turning point around the year 2000.

II. Dataset

A. Source and Scope

Our data is the public "550K+ Spotify Songs (Audio, Lyrics, and Genres)" dataset [1], distributed as two CSV exports named artists.csv and songs.csv, which we used the latter one directly. Each record pairs a track's Spotify audio features with its full lyrics, fetched from LRCLIB, and a main genre mapped to one of ten standard categories. For each song the dataset provides ten numeric audio descriptors (danceability, energy, loudness, speechiness, acousticness, instrumentalness, liveness, valence, tempo, and duration), the lyrics, the release year, genre and sub-genres, popularity, and artist metadata. We analyze all ten audio features together with the lyrics, which carry our two complementary views of mood: how a song *sounds* and what it *says*.

We restricted our data from 1990 through 2022 because the sample size was too small to make a healthy analysis before 1990 and after 2022. **489,761** songs is what we were left with. Coverage is still uneven across the window.

B. Cleaning and Filtering

`songs.csv` is read with Spark in with quote and escape handling enabled so that the commas and line breaks embedded in lyrics are parsed correctly rather than splitting records. Then we apply our three cleaning steps. We first filter out songs with a release year outside of 1990 and 2022, then we filter out the ones with an invalid valence, which is below zero or above one, and lastly, we remove songs without lyrics, genre, year, track identifier, popularity, or sub-genre. The lyrics are normalized in two different ways for two different tasks. For the love, lust and loneliness lexicon-based analysis, we flatten the lyrics to a single white-space normalized string. For BERT analysis, it is important to fit into the token limit, process efficiently, and preserve the meaning. So for BERT, we preserve the line breaks and split the lyrics into eight-line chunks, which are then de-duplicated before ingress.

III. Methodology

A. System Architecture and the Spark Pipeline

The analysis is made as a single Apache Spark job that runs locally on all available cores (`local[7]`). The pipeline works in five stages: (i) load the raw export; (ii) clean it (Section II); (iii) aggregate each audio feature by year; (iv) cluster years into eras; and (v) classify lyric emotion with BERT and persist a dominant emotion per song and then analyze yearly. The cleaned data frame is reused by every analyzer, so it is cached with the `MEMORY_AND_DISK` storage level once, right after filtering.

BERT analysis is expensive; to speed it up we found two solutions. First, lyric chunks are de-duplicated before ingress, yet the count of how many times it has been repeated is kept, so the emotional data is not lost. Second, output is *checkpointed per year*: each year is written to its own partition with a `_SUCCESS` marker, so if a job fails, or is interrupted for any reason, it can continue where it has left off without losing progress.

B. Audio-Feature Aggregation

For each of the ten audio features we compute its mean over all songs released in a given year, outputting a 33-year time series of features.

C. Lyric Emotion Classification with BERT

Lyric emotion is made with the `bert_sequence_classifier_emotion` model from Spark NLP, a BERT sequence classifier fine-tuned to five classes: joy, sadness, anger, love, and fear. Due to lyrics being long and exceeding the token limit of the model, we separate each song's lyrics into eight-line *chunks*. The BERT classifier model has a maximum sentence length of 128 tokens and a batch size of 32. Every chunk gets labeled by one of the five emotions based on its highest probability, and after every chunk's highest probability emotion is found and added up with other chunks' emotion counts c_e for $e \in \{\text{joy, sadness, anger, love, fear}\}$, the song's dominant emotion is determined by the highest count, $\hat{e} = \arg\max_e c_e$. To ensure ties are broken fairly, when k emotions share the highest count we do not award the whole song to a single emotion; instead each tied emotion receives an equal vote of $1/k$. Every song therefore contributes a total weight of exactly one to its year, so the yearly emotion shares still sum to one while no emotion is favored by an arbitrary ordering.

D. Lexicon-Based Theme Detection

We also were curious about how the frequency of the words we associated with love, lust and loneliness changed over the years. we created three categories of words for each and counted their occurances in lyrics of every song, and added up by year and divided the number by how many songs were released that year.

E. K-Means

A question arose in our minds: if humans were reporting worse mental health and dissatisfaction with their lives [2], how was music changing? For answers, we did a K-Means clustering to see how years grouped together based on their valence and energy axes. Each dimension is standardized to zero mean and unit variance before clustering so that features on different scales contribute equally. The number of clusters isn't fixed; we run K-Means for $K = 2, \ldots, 10$, record the within-set sum of squared errors (WSSSE) for each, and select the elbow automatically with the Kneedle heuristic.

F. Statistical Analysis

For every type of analysis, results are a time series of 33 years. Our concern is whether the analysis of audio features or BERT emotions analysis shows a trend and is there a particular year where multiple features shift.

1) Trend Testing: We use Mann-Kendall test, a standard non-parametric trend test. It uses every pair of years and counts how often the next year has the higher value. The result is summarized by Kendall's τ, a number between -1 and $+1$ (-1 = always falling, $+1$ = always rising, 0 = no trend), together with a p-value for significance. We measure the *size* of a trend with Sen's slope, the median yearly change across all pairs of years, which is robust to outliers. We control the false-discovery rate with the Benjamini–Hochberg correction and report a trend as significant only if it survives this stricter bar ($\alpha = 0.05$).

2) Change-Point Detection: We look for sharp turns in the time series. For this, we applied BIC-based method identifying the optimal number of change points. We also applied Pettitt's test to find the most important change point in the series.

IV. Results

A. Audio-Feature Trends

Within the ten audio features, five show statistically meaningful results after Benjamini-Hochberg correction at $\alpha = 0.05$ (Table I, Fig. 1). The strongest decline we see is in valence (Kendall's $\tau = -0.82$, $p < 10^{-9}$), which falls by about 22% compared to its beginning, 1990, level. Acousticness follows valence by a 29% decline. We see increases in loudness by roughly 3.7 dB, energy by about 18%, and tempo by about 4.9 beats per minute. The music over 33 years seems to have become louder, less acoustic, more energetic, faster, yet significantly less positive. Also, there seems to be a steady decrease in duration with a Pettitt change point in 2016.

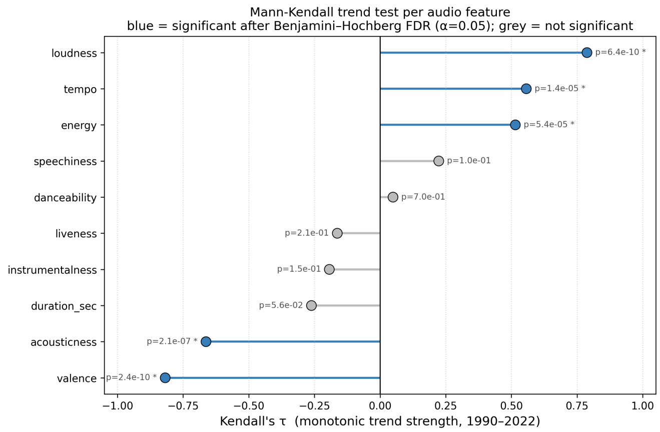

Fig. 1. Mann–Kendall trend strength per audio feature.

TABLE I
MANN–KENDALL TREND TESTS PER AUDIO FEATURE, 1990–2022. ASTERISKS MARK TRENDS SIGNIFICANT AFTER BENJAMINI–HOCHBERG FDR CORRECTION ($\alpha = 0.05$).

Feature	**Kendall τ**	**p (FDR)**	**Direction**
valence	-0.82	2.4×10^{-10}	decreasing*
loudness	$+0.79$	6.4×10^{-10}	increasing*
acousticness	-0.66	2.1×10^{-7}	decreasing*
tempo	$+0.56$	1.4×10^{-5}	increasing*
energy	$+0.52$	5.4×10^{-5}	increasing*
duration	-0.26	0.056	decreasing
speechiness	$+0.22$	0.100	increasing
instrumentalness	-0.19	0.147	decreasing
liveness	-0.16	0.209	decreasing
danceability	$+0.05$	0.698	increasing

B. *Lyric-Emotion Trends Over Time*

In our analysis it seems to be that joy is the dominant emotion throughout the years, showing no significant trend. However, the other four emotions all do. Love goes from 3.3% in 1990 to 2.2% in 2022, anger falls from 21.1% to 18.1%, sadness rises from 21.1% to 23.4%, and fear rises from 2.3% to 2.5%. Four of the five emotions are significant after FDR correction.

C. *Lyric Themes: Love, Lust, and Loneliness*

For every category, we have a list of words we look for, case insensitively. The *love* list is {heart, love, loved, loving, honey, stay, angel, trust, promise, forever, miracle, darling, beautiful, cute}; the *lust* list is {body, bodies, kiss, kissing, kissed, touch, touching, touched, lip, lips, pussy, taste, tasted, tasting, fuck, fucked, fucking}; and the *loneliness* list is {lonely, loneliness, alone, cold, nobody, empty, silence, silent, lost, numb, isolated}. For each song we count how many words fall in each list, then average these counts for each year.

This analysis seems to complement the BERT love results. The density of love per song is U-shaped: high in the early 1990s (about 2.9 love words per song), falling in the mid-2000s (about 2.4), and recovering to its peak by 2022 (about 3.2). The Pettitt analysis shows us that the year 2000 was the single breaking point. Lust seems to have 2013 as its breaking point by the Pettitt analysis, and loneliness's is 2010, increasing after it.

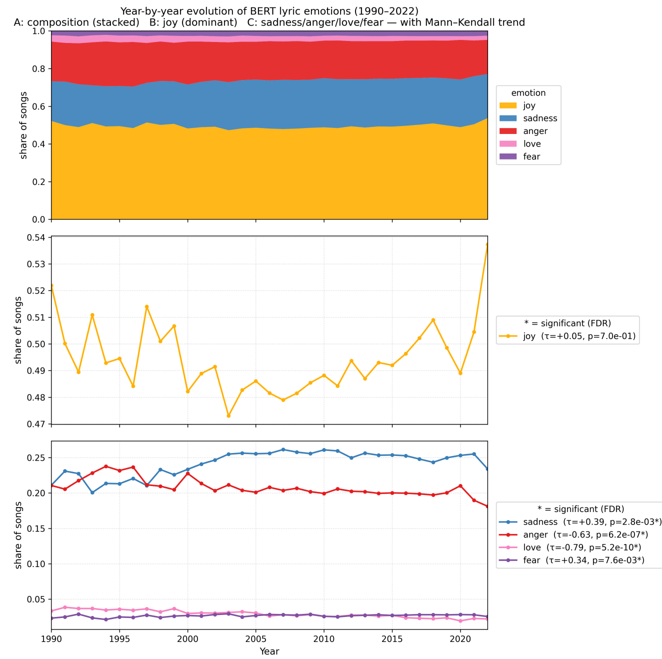

Fig. 2. Year-by-year evolution of the five BERT lyric emotions, 1990–2022.

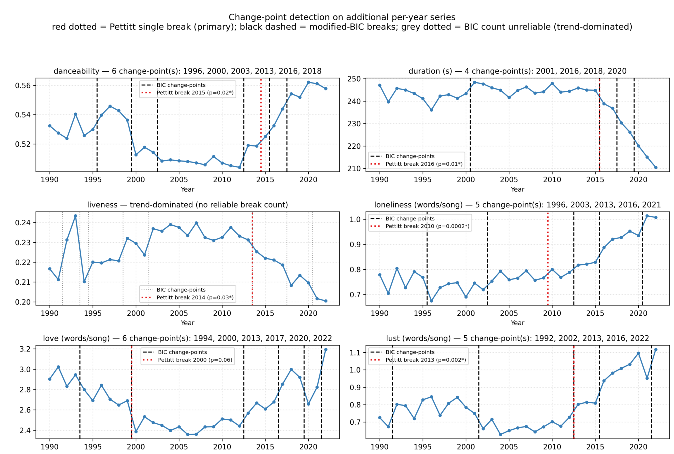

Fig. 3. Per-series change points (BIC-selected breaks and Pettitt single break), including love, lust, and loneliness word usage per song.

D. *Discovered Eras: Mood Clusters*

We used Kneedle heuristic to select $K = 5$ clusters for valence and energy clustering. (Fig. 6): There seems to be a steady fall in valence and a rise in energy that peaks in the late 2000s and 2010.

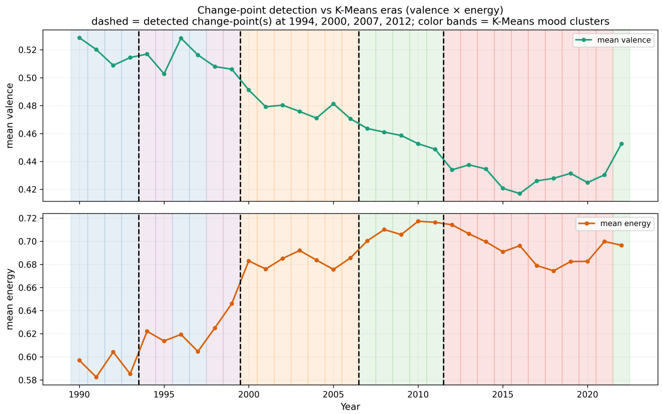

Fig. 4. Mean valence and energy over time with BIC-selected change-points (dashed) and K-Means mood-cluster bands (background).

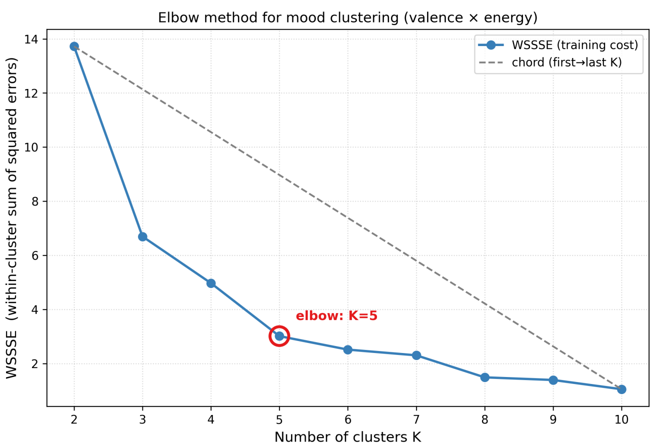

Fig. 5. WSSSE versus K for the mood clustering, with the Kneedle-selected elbow at $K = 5$.

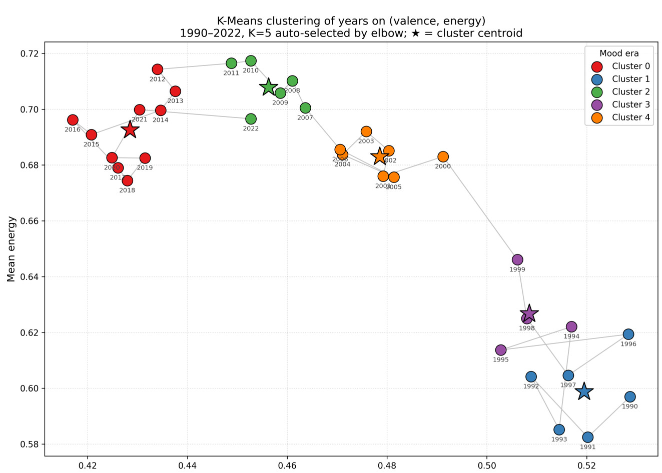

Fig. 6. K-Means clustering of years on (valence, energy).

V. Discussion

A. Interpretation of the Observed Trends

The analyses show that sonically songs became louder, faster, more energetic, and less acoustic over the 33-year period, while the lyrics became less dominated by love and more by sadness and fear, although joy remains as the dominant emotion in lyrics. The most interesting finding is that the clustering of years based on valence and energy shows us that similar years group together, and in a society where people are reporting more anxiety and less life satisfaction, the musical analysis shows that the valence is lowering while energy is increasing. Also, the COVID-19 period and its aftermath show some interesting results, although we cannot claim any causal relationship, the correlation exists. Around 2020, the valence, love word count decreased with the BERT love and happiness share while anger, sadness, and fear increasing. According to our lexicon analysis, 2021 is a BIC change point for loneliness.

B. Threats to Validity and Limitations

Several limitations have been found. First, the corpus is a sample from the Spotify, and real people have contributed to it by using the Spotify's API, and the songs weren't randomly selected to be put into the dataset provided. Second, the BERT emotion classifier model has been trained on general text, not song lyrics, which is way more complex with emotional expression, nuances and figurative language, which the model may have interpreted wrongly, and has fixed weights, so fine tuning is not possible. We have chosen 50 randomly selected songs and fed them into Claude Opus 4.8 Max and asked it to classify the lyrics into the five emotions, and compared the results. The LLM agent and BERT emotion model has disagreed in 32 out of 50 songs, with ties in the BERT chunk counts broken by the model's own summed confidence rather than an arbitrary ordering. The most disagreements arose from emotion joy, which Claude labeled as sadness (8 times), or love (8 times). It seems like the BERT model over-predicts joy. Third, we do not report any causal relationship between results and external events, we only report their correlation.

References

[1] serkantysz, "550K+ Spotify Songs (Audio, Lyrics, and Genres)," Kaggle, 2024. [Online]. Available: https://www.kaggle.com/datasets/serkantysz/550k-spotify-songs-audio-lyrics-and-genres

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